

Technical Design Document: Identity Document Fraud Detector

1. Executive Summary

This system is a hybrid computer vision and deep learning pipeline designed to distinguish between bona-fide (genuine) identity documents and fraudulent/counterfeit presentation attacks.

The system specifically targets two common fraud vectors:

Digital Manipulations: Generative AI edits or pixel-level document alterations.

Physical Presentation Attacks (Print-and-Capture): Physical document printouts re-photographed to bypass automated checks, which typically bypass digital-only watermark checks.

2. System Architecture

The pipeline is structured into three distinct, decoupled phases:

Phase 1: Metadata & Frequency Feature Extraction

Scans raw image directories and compiles structural and mathematical signatures into a JSON payload (true_specs.json and false_specs.json).

Fast Fourier Transform (FFT): Converts the image into the frequency domain to isolate unnatural periodic structures (such as Moiré patterns from digital screens or halftone printing grids).

Color Channels: Extracts color distribution shifts (RGB channel means and standard deviations).

Phase 2: Deep Learning Classifier

Trains a lightweight, high-performing convolutional network on labeled datasets.

Backbone: EfficientNet-B0 (pre-trained on ImageNet).

Output: Custom single-node Sigmoid head returning a continuous real-valued probability score range in $[0, 1]$, where 1.0 represents a high-probability attack.

Phase 3: Physical vs. Digital Inference

Processes an unseen target folder (test_documents) to produce a final output.csv. It classifies the document type, extracts geographical metadata via file-naming schemas, and runs a heuristic "analog hole" check to determine if the document is natively digital or physically re-captured.

3. Core Mathematical Formulas

Laplacian Variance (Digital-Native vs. Re-capture Verification)

To bypass physical print-and-capture fraud, we evaluate high-frequency edge degradation using the variance of the Laplacian of the image (l):

Read: https://github.com/neckleshole-source/The-Freuid-Challenge-2026---IJCAI---ECAI_competition/blob/main/README.md

Performance Evaluation: The FREUID Metric

The model's operational readiness is validated using the custom FREUID Score (where lower is better).

This metric balances global model quality with a strict operational security threshold.

1. Global Quality Score (g_{audet}): Based on the Area under the Detection Error Trade-off curve (AuDET),

2. Operational Quality Score (g_{apcer}): Measures the False Negative Rate (APCER) at a highly strict 1% False Positive Rate (BPCER) limit ,

3. Harmonic Mean Optimization:

Read: https://github.com/neckleshole-source/The-Freuid-Challenge-2026---IJCAI---ECAI_competition/blob/main/README.md

4. Input & Output Schema Map

Training Input Data Requirements

Images: Standard image formats (.png, .jpg, .jpeg).

CSV Labels: Two-column file (image_name, label), where 0 for genuine, 1 for attack

Exported System Outputs

1. Metadata Specs (true_specs.json / false_specs.json):

```
```text
{
 "filename": "US_DL_sample.jpg",
 "aspect_ratio": 1.586,
 "blurriness_score": 240.15,
 "frequency_fft_mean": 45.12
}
```
```

Production Inference (output.csv):

- image id: Unique file identifier.
- label: Predicted probability score [0.0, 1.0].
- is_digital: Boolean indicator (1 = \text{digital-native}, 0 = \text{physical re-capture}).
- type: Parsed identity type (e.g., US_DL, DE_ID).

5. Execution Instructions

To run the end-to-end pipeline locally or inside Kaggle:

1. Populate ./true_documents and ./false_documents with your respective dataset files.
2. Ensure true_labels.csv and false_labels.csv are in the project root.
3. Run the pipeline script:

```
```text
python identity_detector.py
```
```